- Find the allowable values of E that permit a mathematical solution.
- Find the eigenvectors and possible eigenvalues of the energy.

This is a little trickier than you might think. There turns out to be a solution to the equation for every value of E, including all the complex numbers, but most solutions are physically absurd. If we just start at some point and solve the Schrödinger equation by making little incremental steps, we will almost always find that $\psi(x)$ grows or "blows up" as x becomes large. In other words, we may be able to find solutions to the equation, but only very rarely will we find a normalizable solution.

In fact, for most values of E, including all the complex numbers, the solutions of Eq. 10.14 grow exponentially as x approaches ∞, $-\infty$, or both. This type of solution makes no physical sense; it tells us that there is an overwhelming probability that the oscillator coordinate is infinitely far away. Clearly, we want to impose some condition that gets rid of such solutions. So let's impose one:

Physical solutions of the Schrödinger equation must be normalizable.

This is a very powerful constraint. In fact, for almost all values of E, there are no normalizable solutions. But for certain very special values of E such solutions do exist, and we will find them.

10.5 THE GROUND STATE

What is the lowest possible energy level for a harmonic oscillator? In classical physics, the energy can never be negative because the Hamiltonian has an x^2 term and a p^2 term; to minimize energy, we just set p and x equal to zero. But in quantum mechanics, that's asking too much. The uncertainty principle says that you *can't* set both x and p equal to zero. The best you can do is find a compromise state in which x and p are not too spread out. Because you have to compromise, the lowest possible energy will *not* be zero. Neither p^2 nor x^2 will be zero. Because the operators $\mathbf{X}^2$ and $\mathbf{P}^2$ can